# WEEK#06a

## BIOS & DOS INTERRUPT SERVICES

Develop understanding to use different BIOS and DOS Interrupt Services using MASM/TASM utility.

### I. What is Interrupt?

Interrupt is considered as a signal, which is generated when I/O function is required, to get the attention of the CPU. For example, mouse click or key press is considered as **hardware interrupts**, whereas, a disk I/O requested by a program is considered as **software interrupts**.
Interrupt causes the CPU to halt the currently executing program. It is somewhat like a FAR CALL. When it is invoked, it saves the value of **CS:IP** for the next instruction to be executed and **flags** on the stack and load the value of the **CS:IP** of the **ISR** (Interrupt Service Routine), associated with the interrupt, from the **IVT** (Interrupt Vector Table). The procedure is depicted in Figure 5.1.
The **IVT** is shown in Figure 5.2. It is the pointer associated with an **interrupt** type code and the subroutine (or procedure) being written to serve the **interrupts** associated with that code.
There are some extremely useful subroutines within BIOS and DOS that are available to the user through the INT (Interrupt) instructions. The **8086/88** supports total 256 types i.e. 00H to FFH. The INT instruction has the following format

**INT xx ;the interrupt number xx can be 00 to FFh**

This makes 256 interrupts in 80x86 microprocessors. Of these 256 interrupts, two are widely used namely, INT 21H (DOS) and INT 10H (BIOS). Each of them can perform number of functions. Various functions of INT 21H and INT 10H are selected by the value put in the AH register. Other registers are used to pass parameters and data to sub-function as and where necessary. Some of the functions are given below with registers other than AH to pass the necessary parameters.

- To display character → function 2 with INT 21h
  *DL = character, AH = 2, INT 21h*
- To display string → function 9 with INT 21h
  *DX = offset of string, AH = 9, INT 21h*
- To clear the screen → function 6 with INT 10h
  *AL = 0, BH = 7, CX = top left window coordinates,*
  *DX = bottom right window coordinates, AH = 6, INT 10h*
- To set the cursor → function 2 with INT 10h
  *BH = 0 (page 0), DL = column position, DH = row position,*
  *AH = 2, INT 10h*

*Download and bring the documents regarding **INT 10h** and **INT 21h** from the course website.*
https://ssuet.org/course/view.php?id=1415#section-6

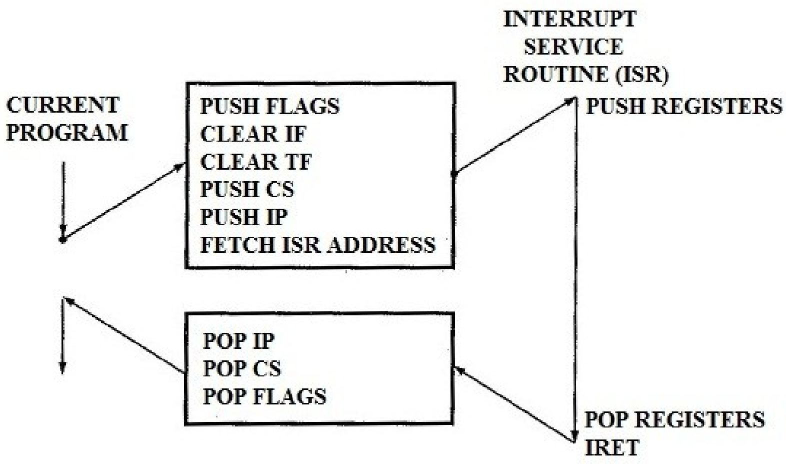


**Figure 5.1 The 8086/88 Interrupt Procedure**

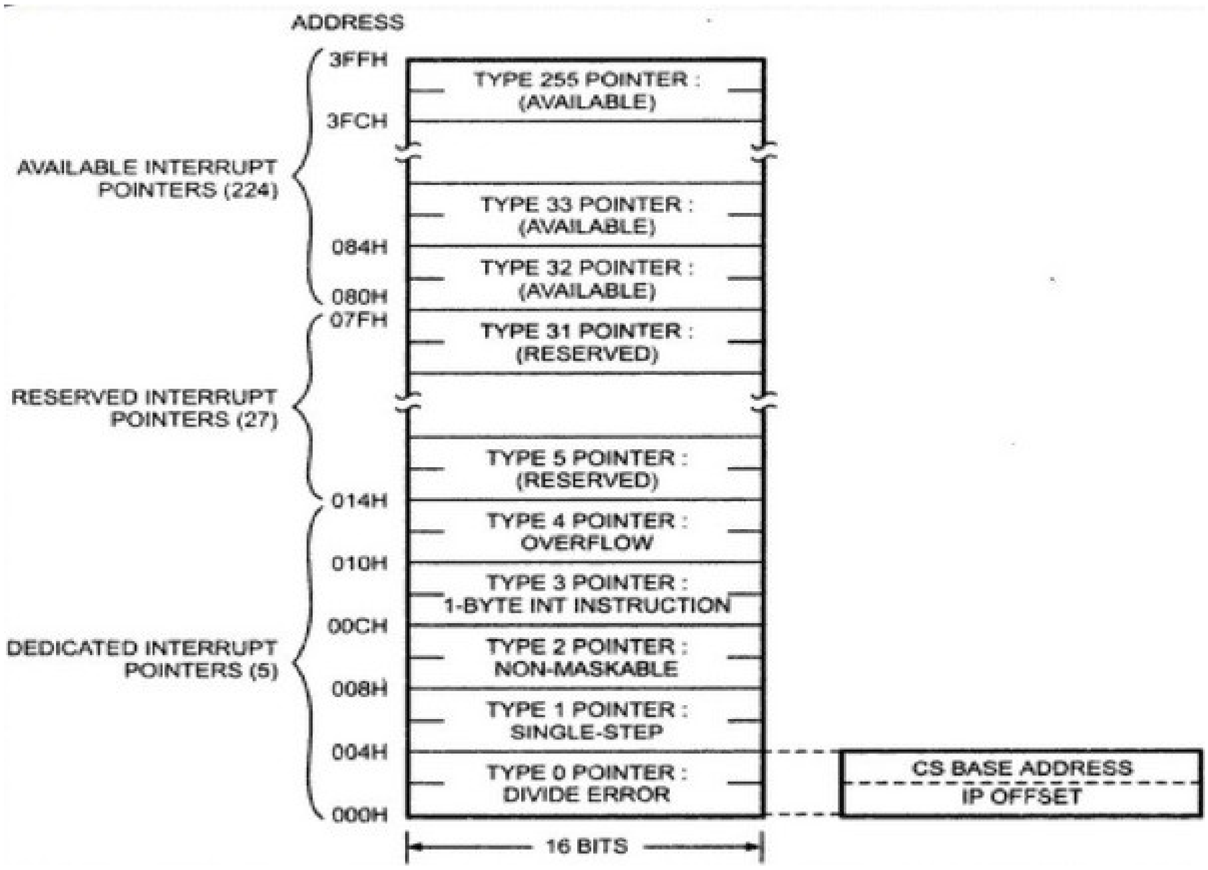


**Figure 5.2 The 8086/88 Interrupt Vector Table (IVT)**